

Development and external validation of a clinical prediction model for new-onset atrial fibrillation in intensive care: a multicentre, retrospective cohort study



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Summary

Background New-onset atrial fibrillation, a condition associated with adverse outcomes in the short and long term, is common in patients admitted to intensive care units (ICUs). Identifying patients at high risk could inform trials of preventive interventions and help to target such interventions. We aimed to develop and externally validate a prediction model for new-onset atrial fibrillation in patients admitted to ICUs.

Methods We conducted a multicentre, retrospective cohort study in three ICUs across the UK and four ICUs across the USA. Patients aged 16 years and older admitted to an ICU for more than 3 h without a history or presentation of clinically significant arrhythmia were eligible for inclusion. We analysed clinical variables to investigate the associations between predetermined candidate variables and risk of new-onset atrial fibrillation and to develop a model to estimate these risks. We developed the METRIC-AF model, a machine learning model incorporating dynamic variables. Model performance was assessed through internal–external cross-validation during model development and externally validated by use of multicentre data from ICUs across the UK. We then developed a simple graphical prediction tool using three important predictors.

Findings Among 39 084 eligible patients admitted to an ICU between 2008 and 2019, 2797 (7.2%) developed new-onset atrial fibrillation during the first 7 days of their ICU stay. We identified multiple non-linear associations between candidate variables and risk of new-onset atrial fibrillation, including hypomagnesaemia at serum concentrations below 0.70 mmol/L. The final METRIC-AF model contained ten routinely collected clinical variables. Compared with a published logistic regression model, the METRIC-AF model showed superior calibration, net benefit across clinically relevant risk thresholds, and discriminative performance (C statistic 0.812 [95% CI 0.805–0.822] vs 0.786 [0.778–0.801]; $p=0.0003$). The simple graphical tool performed well in attributing the risk of new-onset atrial fibrillation in the external validation dataset (C statistic 0.727 [95% CI 0.716–0.739]).

Interpretation The METRIC-AF model and its companion graphical tool could support the identification of patients at increased risk of developing new-onset atrial fibrillation during ICU admission, informing targeted prophylactic strategies and trial enrichment by use of routinely available clinical data. An online app also developed as part of the study allows for the exploration of prediction generation among individuals and external validation in prospective studies.

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Introduction

Atrial fibrillation is a common arrhythmia in patients admitted to an intensive care unit (ICU).¹ New-onset atrial fibrillation during critical illness is associated with increases in length of ICU and hospital stay, risk of in-hospital mortality,² and health-care costs.³ A causal relationship between new-onset atrial fibrillation and adverse ICU outcomes is unproven, but its onset in ICUs has been associated with subsequent reductions in blood pressure and increased use of vasoactive medications.⁴ The condition also has a long-term burden, with patients who develop new-onset atrial fibrillation in an ICU at higher risk of readmission to hospital with atrial fibrillation, heart failure, and stroke than those who do not, independent of

pre-existing comorbidities.^{5,6} Given these associations, the prevention of new-onset atrial fibrillation during critical illness might improve patient outcomes.

Several atrial fibrillation prediction models have been developed for use outside of hospital.⁷ Risk factors for the development of the condition in this setting differ from those of critically ill, hospitalised patients. Age and comorbidities still contribute to risk;⁸ however, the severity of acute illness and some interventions (eg, vasoactive medications) also have a key role.⁹ Furthermore, although community risk models are built on data spanning years, the prediction of atrial fibrillation during critical illness would be relevant over timeframes of hours to days. These differences make models of atrial fibrillation risk in the

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Research in context

Evidence before this study

New-onset atrial fibrillation is common in patients treated in intensive care units (ICUs) and is associated with adverse outcomes in the short and long term. Identifying patients at high risk could inform trials of preventive interventions and help to target such interventions. In our published review, we searched MEDLINE, Embase, Cochrane Database of Systematic Reviews, and CENTRAL for articles published between Jan 1, 1970, and Aug 30, 2018. We excluded non-English-language papers where no translation was available. We included studies of adults based in general ICUs that evaluated predictors for new-onset atrial fibrillation. We excluded studies involving patients with a history of atrial fibrillation. This systematic review of risk factors for new-onset atrial fibrillation in ICU patients showed one prediction model for use in this context. Since then, few models for predicting new-onset atrial fibrillation have been developed for critically ill patients, and are limited by poor performance in external validation, the inclusion of predictors that are not routinely collected, and inadequate assessment of model calibration. Existing models are specific to individual health-care systems and not largely generalisable to other contexts; therefore, a broadly applicable and clinically useful tool for predicting the risk of new-onset atrial fibrillation in patients admitted to an ICU is required.

Added value of this study

The METRIC-AF model is a novel prediction tool based on machine learning that is designed to estimate the risk of

new-onset atrial fibrillation in patients admitted to an ICU by the use of ten routinely collected clinical variables. The model was developed with large multicentre datasets, including 39 084 patients from the UK and the USA, and externally validated in a cohort of 33 844 patients from the UK. Compared with a previously published logistic regression model, the METRIC-AF model showed superior discrimination and net benefit across clinically relevant risk thresholds. Furthermore, analysis of individual predictors revealed the novel finding that hypomagnesaemia (serum concentration <0.7 mmol/L) was associated with an increased risk of new-onset atrial fibrillation in the general ICU setting.

Implications of all the available evidence

The METRIC-AF model and its simplified tool (a graphical tool with three predictors) include objective and readily available predictors and can be used at a patient's bedside to stratify patients by risk of new-onset atrial fibrillation. The model incorporates multicentre data from the UK and the USA, potentially improving generalisability over models developed in single centres. An improved understanding of individual predictors and the identification of patients at high risk of new-onset atrial fibrillation could inform targeted trials of preventive interventions to improve outcomes for this patient group.

community unsuitable for the prediction of new-onset atrial fibrillation during acute illness in hospital settings.

A few prediction models of new-onset atrial fibrillation have been developed for critically ill patients in hospital, but they are limited by poor performance in external validation cohorts, the inclusion of predictors that are not routinely collected, and inadequate assessment of model calibration.^{10–12} Existing models are specific to health-care systems and not largely generalisable; therefore, a broadly applicable and clinically useful tool for predicting the risk of new-onset atrial fibrillation is needed. Applying machine learning methods might improve prediction performance in the context of new-onset atrial fibrillation by capturing complex variable interactions that are often overlooked in regression models.¹³ We aimed to develop and externally validate a prediction model for new-onset atrial fibrillation in patients admitted to an ICU.

Methods

Study design

We conducted a multicentre, retrospective cohort study using data from two large ICU databases from the UK (PICRAM¹⁴) and the USA (MIMIC-IV [version 1.0]¹⁵). The PICRAM database comprises data relating to more than 12 000 patients treated in three general ICUs in Oxford and Reading (UK) from 2008 to 2015.¹⁴ The MIMIC-IV database

comprises data from over 50 000 patients admitted to critical care units at a tertiary care hospital in Boston (MA, USA) between 2008 and 2019.¹⁵

We combined data from three ICUs in the PICRAM database and four ICUs in the MIMIC-IV database to develop the model. We used the Critical Care Health Informatics Collaborative (CCHIC) database¹⁶ for external model validation. The CCHIC database comprises data from over 33 000 patients admitted to ICUs in four health-care trusts in London and Cambridge (UK) from 2014 to 2018.

Ethical approval was received, with informed consent waived for all cohorts involved (PICRAM 11/SC/0440, MIMIC-IV 2001P001699, and CCHIC 14/LO/1031). The PICRAM study was granted ethical approval by the National Research Ethics Service Committee Oxford (11/SC/0440) and the National Information Governance Board (ECC 7–05(f)/2011). Use of patient data without informed consent was approved by the National Information Governance Board on Feb 2, 2012 (ECC 7–05(f)/2011). The use of these data for this work was approved by the University of Oxford Critical Care Research Group Data Access Committee on Sept 20, 2019. The MIMIC-IV database and its associated use was approved by the institutional review boards of Beth Israel Deaconess Medical Center (Boston,

MA, USA) and the Massachusetts Institute of Technology (Cambridge, MA, USA; 2001P001699).

The legal basis for transferring data was provided to those who created the CCHIC dataset under section 251 of the National Health Service Act 2006 (Confidentiality Advisory Group reference 14/CAG/1001). Ethical approval was granted by a Health Research Authority Research Ethics Committee (14/LO/1031). Use of the CCHIC database (Internal Collaborator Release 2018-09-14) for this work was registered and approved by the CCHIC scientific advisory group in September, 2018.

The statistical analysis plan has been published previously;¹⁷ changes to this plan are outlined in the appendix (p 2). We report this study according to TRIPOD+AI guidelines.¹⁸

Participants

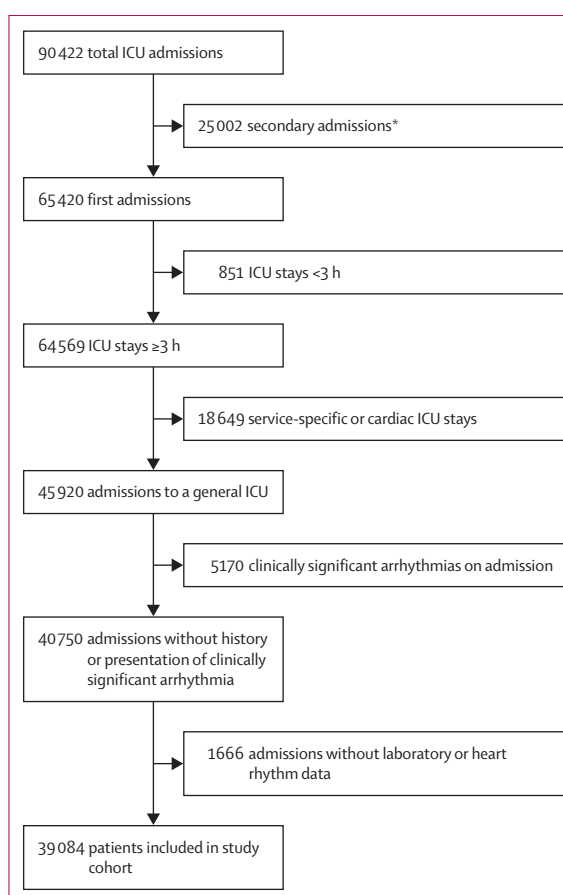
Patients aged 16 years and older who were admitted to one of the study ICUs for more than 3 h without a history or presentation of clinically significant arrhythmia were eligible for inclusion. For patients admitted more than once to a study ICU, we used their first admission. Patients who were cared for by a coronary care or cardiac surgery team, had an ICU length of stay of less than 3 h, had significant arrhythmia in the first 3 h of arrival to the ICU as a surrogate for pre-existing undocumented arrhythmia, or had documented pre-existing arrhythmias were excluded.

Procedures

Routine practice in all included ICUs was to display a three-lead electrocardiogram continuously, with heart rhythm recorded by the bedside nurse at regular intervals. We defined atrial fibrillation as the documentation of atrial fibrillation lasting for at least 30 s in the patient observation chart.¹⁹ A documented heart rhythm was assumed to persist until the next identifiable rhythm was recorded. Pre-existing arrhythmia was defined as any entry of clinically significant arrhythmia in the patient's medical history or any record of medications prescribed with an indication of heart rhythm management in their medication history. New-onset atrial fibrillation was defined as atrial fibrillation occurring after 3 h of admission in a patient with no documented pre-existing arrhythmia. Data were restricted to the first 7 days of each ICU admission.

We conducted an international Delphi consensus exercise to identify candidate predictors of new-onset atrial fibrillation during critical illness. This involved a three-round process with subject experts to refine and prioritise key variables.²⁰ Full details of predictor variable handling and candidate predictors are provided in the appendix (pp 1, 22).

Given that sample size calculations for machine learning methods are not well established, the sample size was calculated based on the recommended approach for regression-based prediction models.²¹ Assuming an outcome proportion of 7.5%,⁴ 67 predictor parameters (47 candidate predictors, of which ten required three parameters for restricted cubic splines), and an anticipated C statistic of 0.80,¹⁰ the minimum sample size for new



See Online for appendix

Figure 1: Cohort profile for the MIMIC-IV and PICRAM databases combined ICU=intensive care unit. *For patients admitted more than once to a study ICU, only data from their first admission was used.

model development was estimated to be 6496 patients (487 new-onset atrial fibrillation events).²²

We analysed data missingness and plotted the missingness of variables (patients with >1% missing data) within 3-h intervals over time. We used the last observation carried forward to reflect how the resulting models would be used in clinical practice (ie, a clinician using the most recent value available to them).²³ Where no data were available within a 3-h period nor available to be carried forward, these datapoints were imputed by use of a modelling approach based on a multivariable normal distribution.²⁴ The imputation model included the outcome variable and all variables considered for the multivariable model building. The outcome carried information about the missing values of the predictors.²⁵ Therefore, including the outcome in the imputation model preserved the relationships between predictors and outcomes, ensuring unbiased estimates of association.²⁶ Outcomes were not imputed. A single imputation was used due to limitations with pooling across imputations with the use of machine learning models.

We explored unadjusted associations between candidate predictors and risk of new-onset atrial fibrillation using logistic regression, allowing for non-linear associations

	No atrial fibrillation (n=36 287)	New-onset atrial fibrillation (n=2797)
Age, years	61 (48-73)	74 (65-82)
Sex		
Female	16 579 (45.7%)	1241 (44.4%)
Male	19 708 (54.3%)	1556 (55.6%)
Ethnicity		
Asian	5623 (15.5%)	560 (20.0%)
Black	3167 (8.7%)	151 (5.4%)
Hispanic	1111 (3.1%)	37 (1.3%)
White	20 073 (55.3%)	1596 (57.1%)
Other	2137 (5.9%)	119 (4.3%)
Unknown	4176 (11.5%)	334 (11.9%)
Weight, kg	76 (65-90)	76 (65-91)
Chronic obstructive pulmonary disease	6674 (18.4%)	737 (26.3%)
Hypertension	16 359 (45.1%)	1675 (59.9%)
Congestive cardiac failure	4360 (12.0%)	796 (28.5%)
Ischaemic heart disease	6063 (16.7%)	839 (30.0%)
Admission year		
2008-10	11 810 (32.5%)	906 (32.4%)
2011-13	10 971 (30.2%)	832 (29.7%)
2014-16	9504 (26.2%)	712 (25.5%)
2017-19	4002 (11.0%)	347 (12.4%)
Invasive mechanical ventilation	10 427 (28.7%)	1080 (38.6%)
Presence of central venous catheter	6473 (17.8%)	761 (27.2%)
FiO ₂	0.41 (0.33-0.60)	0.50 (0.37-0.81)
Serum sodium concentration, mmol/L	138.0 (135.2-141.0)	137.7 (134.4-140.3)
Serum potassium concentration, mmol/L	4.0 (3.7-4.5)	4.2 (3.8-4.7)
Serum magnesium concentration, mmol/L	0.78 (0.70-0.86)	0.78 (0.70-0.91)
Serum urea concentration, mmol/L	6.1 (4.3-9.7)	9.6 (6.4-16.1)
Serum creatinine concentration, µmol/L	80 (62-115)	106 (71-177)
Serum albumin concentration, g/L	31 (25-36)	28 (23-33)
White cell count, × 10 ⁹ per L	10.7 (7.6-14.9)	11.6 (7.7-16.3)
Neutrophil count, × 10 ⁹ per L	7.9 (3.8-12.2)	8.8 (4.1-13.8)
Lymphocyte count, × 10 ⁹ per L	0.8 (0.4-1.4)	0.7 (0.3-1.1)
Log neutrophil-to-lymphocyte ratio	9.5 (5.1-17.1)	11.9 (6.2-22.2)
Serum C-reactive protein concentration, mg/L	34 (9-97)	77 (24-145)
Haemoglobin concentration, g/L	111 (95-126)	106 (92-122)
Platelet count, × 10 ⁹ per L	203 (147-269)	191 (132-257)
Prothrombin time, s	14.2 (12.4-16.2)	15.5 (13.3-18.8)
Serum lactate concentration, mmol/L	1.7 (1.1-2.6)	1.9 (1.3-3.1)
Blood pH	7.37 (7.31-7.42)	7.34 (7.27-7.40)
Peripheral oxygen saturation, %	96.0 (94.0-99.0)	95.0 (93.0-98.0)
Mean arterial pressure, mm Hg	71 (62-82)	66 (58-76)
Heart rate, beats per min	92 (79-107)	93 (80-109)
Respiratory rate, breaths per min	21 (18-26)	23 (19-28)
Body temperature, °C	36.8 (36.4-37.2)	36.7 (36.3-37.1)
Presence of atrial ectopic beats	1116 (3.1%)	232 (8.3%)
ICU length of stay, days	1.7 (0.9-3.3)	4.5 (2.4-8.8)
Deaths in ICU	2312 (6.4%)	550 (19.7%)
Hospital length of stay, days	8 (4-14)	12 (7-21)
Deaths in hospital	3853 (10.6%)	834 (29.8%)
Admission ICU		
Oxford General	4357 (12.0%)	494 (17.7%)
Oxford Churchill	2411 (6.6%)	167 (6.0%)
Reading General	2786 (7.7%)	170 (6.1%)

(Table continues on next page)

using restricted cubic splines with two internal knots. To account for the confounding association between new-onset atrial fibrillation and acute kidney injury and serum electrolyte concentrations,²⁷ we performed an analysis adjusted for renal function to explore the association between serum magnesium and potassium concentrations during the ICU admission and risk of new-onset atrial fibrillation. We then fitted a multivariable model, including time-matched serum urea and creatinine concentrations as covariates.

To model dynamic (repeated measures) data, we used values updated at 3-h intervals throughout the patient's ICU stay (the observation window) to predict the subsequent risk of new-onset atrial fibrillation over a rolling 24-h window (the prediction window), excluding the immediately subsequent 3 h from the time of prediction as a buffer (appendix p 5). Only data collected before the first episode of new-onset atrial fibrillation were used for prediction; subsequent recurrence was not modelled.

Model development and validation

We developed a gradient-boosted tree machine learning model. We performed variable selection through recursive feature elimination, aiming to produce a parsimonious model to aid clinical utility and reduce the risk of overfitting. Hyperparameter optimisation was nested within each step of variable elimination. The model was trained by the use of a large grid search of hyperparameters, including learning rate, number of trees, maximum depth, and minimum child weight.

First, for each ICU-level fold with one held-out (ie, not used for model training) ICU (outer cross-validation), the model was trained and hyperparameters were optimised by use of all the predictors on the training data (six ICUs); predictions in the held-out ICU (patient-level data) were generated; and variables were ranked by importance. For each subset of numbers of predictors (S_i), the most important S_i predictors were kept; the model was trained on the training data (six ICUs) with S_i predictors (inner cross-validation); and predictions were generated in the held-out ICU from the outer cross-validation. Second, the average performance was calculated across the seven held-out ICUs. Third, i predictors from the model with the fewest predictors that achieved an area under the receiver operating characteristic within 1% of the model comprising all candidate predictors were selected. Fourth, the final model was fitted on the full dataset with the i predictors selected in the previous step.

Given the multiple nested cross-validation procedures, variable ranking methods were necessarily less intensive and faster than calculation of Shapley values, although potentially less robust. Variable importance was based on the so-called gain statistic, a measure of the reduction in loss brought by a predictor to the model when the data are split based on that predictor.²⁸

We used internal-external cross-validation to explore the heterogeneity in model performance during model development. In this procedure, performance metrics were

averaged over sub-models fitted on each permutation of six ICUs, tested on the remaining held-out ICU. To account for changing risk during an ICU stay, models were trained with data that included repeated measures. To present risks and performance metrics at the patient level, models were tested on patient-level data, achieved through random sampling to select a single observation set per patient (appendix p 5).²⁹ Each patient was resampled five times to obtain a spread of data over their ICU admission. At each iteration of feature elimination, performance statistics were averaged over 35 cross-validation folds (five per patient with seven ICU-level folds; appendix p 6).

We generated Shapley values to explore variable importance (appendix p 1). We used a partial dependence analysis for each continuous variable to show how average model predictions changed over a range of values. We then superimposed a sample of individual conditional expectation curves to show how changes in the variable of interest affected the risk of individual patients in the dataset based on other covariates.

We externally validated the final model on the CCHIC dataset. Data preparation, imputation methods, and outcome definitions were the same as those used for the data in the development of the model. All predictors were available in the external validation dataset, except for the presence of atrial ectopic beats. Values for this predictor were randomly imputed as a binary variable at the baseline prevalence found in the development dataset (3.2%). We performed a sensitivity analysis, in which model discrimination was assessed across 1000 imputations of the missing variable, each with a random prevalence of 0–7% to assess the effects on model performance.

External validation performance was compared with a previously published comparator model: a logistic regression model developed in two ICUs in the Netherlands (the Atrial Fibrillation in Sepsis [SAFE] score).¹⁰ We calculated model discrimination and the calibration slope. We calculated 95% CIs from 100 bootstrap resamples. We then plotted calibration plots of expected versus observed proportions of patients with the outcome of interest (new-onset atrial fibrillation) across ten groups stratified by risk. We then added smoothed calibration lines to the plots for each model by regressing the binary outcome on the predicted probability of the outcome from the fitted model with a loess smoother.³⁰

To promote a fair comparison, we recalibrated the comparator model's intercept and slope (logistic recalibration) to the development data and present calibration curves before and after recalibration. To compare the C statistic values between the METRIC-AF model and the comparator model, a p value was calculated with a permutation test. We then performed a decision curve analysis for the METRIC-AF model and the comparator model to compare each model's net benefit over various risk thresholds. To investigate heterogeneity in model performance,¹⁸ we assessed model discrimination and calibration within subgroups of patients stratified by sex, ethnicity, and age³¹ in the external validation dataset.

	No atrial fibrillation (n=36 287)	New-onset atrial fibrillation (n=2797)
(Continued from previous page)		
Boston Medical	7835 (21.6%)	620 (22.2%)
Boston Medical-Surgical	6785 (18.7%)	500 (17.9%)
Boston Surgical	6612 (18.2%)	430 (15.4%)
Boston Trauma	5501 (15.2%)	416 (14.9%)
Data are median (IQR) or n (%). Category percentages might not sum to 100% due to rounding. FiO ₂ =fractional concentration of oxygen in inspired air. ICU=intensive care unit.		
Table: Patient characteristics at ICU admission		

We created an online prediction app with tools to promote external validation of the model. The METRIC-AF prediction model is available online as a web-based prediction app. To facilitate independent external validation, users can also upload their own dataset to which predicted risks are appended.

As a second approach to improve the bedside utility of the gradient-boosted tree model, we developed a simple graphical tool. We used a partial dependence analysis to generate predictions across a range of the three variables with the highest importance according to the Shapley analysis, while accounting for the true values of the remaining predictors and their interactions (appendix pp 1–2). From these predictions, we created five groups on a graphical tool stratified by risk. We assessed the discriminative performance of the score across all five groups in the external validation dataset and report the proportion of patients with new-onset atrial fibrillation by risk group. We used resampling with replacement (n=1000) to estimate the 95% CIs of these estimates.

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

The development dataset contained data from 39 084 patients admitted to an ICU between 2008 and 2019 (figure 1). 2797 (7.2%) of these patients developed new-onset atrial fibrillation during the first 7 days of their ICU stay. Individual cohort profiles for the MIMIC-IV and PICRAM databases are provided in the appendix (pp 7–8). The proportions of patients included by ICU unit and year of admission are shown in the appendix (pp 9–10). Given that the PICRAM database comprised UK-based ICU data from 2008 to 2015, data from 2017 to 2019 contained patients from ICUs in the USA only.

The CCHIC database contained 33 844 patients with index ICU admissions, of whom 29 833 (88.1%) fulfilled the inclusion criteria (appendix p 8), with 2686 (9.0%) developing new-onset atrial fibrillation during admission. In both model development and external validation datasets, patients who developed new-onset atrial fibrillation

For the app see <https://metric-af.shinyapps.io/metric-af/>

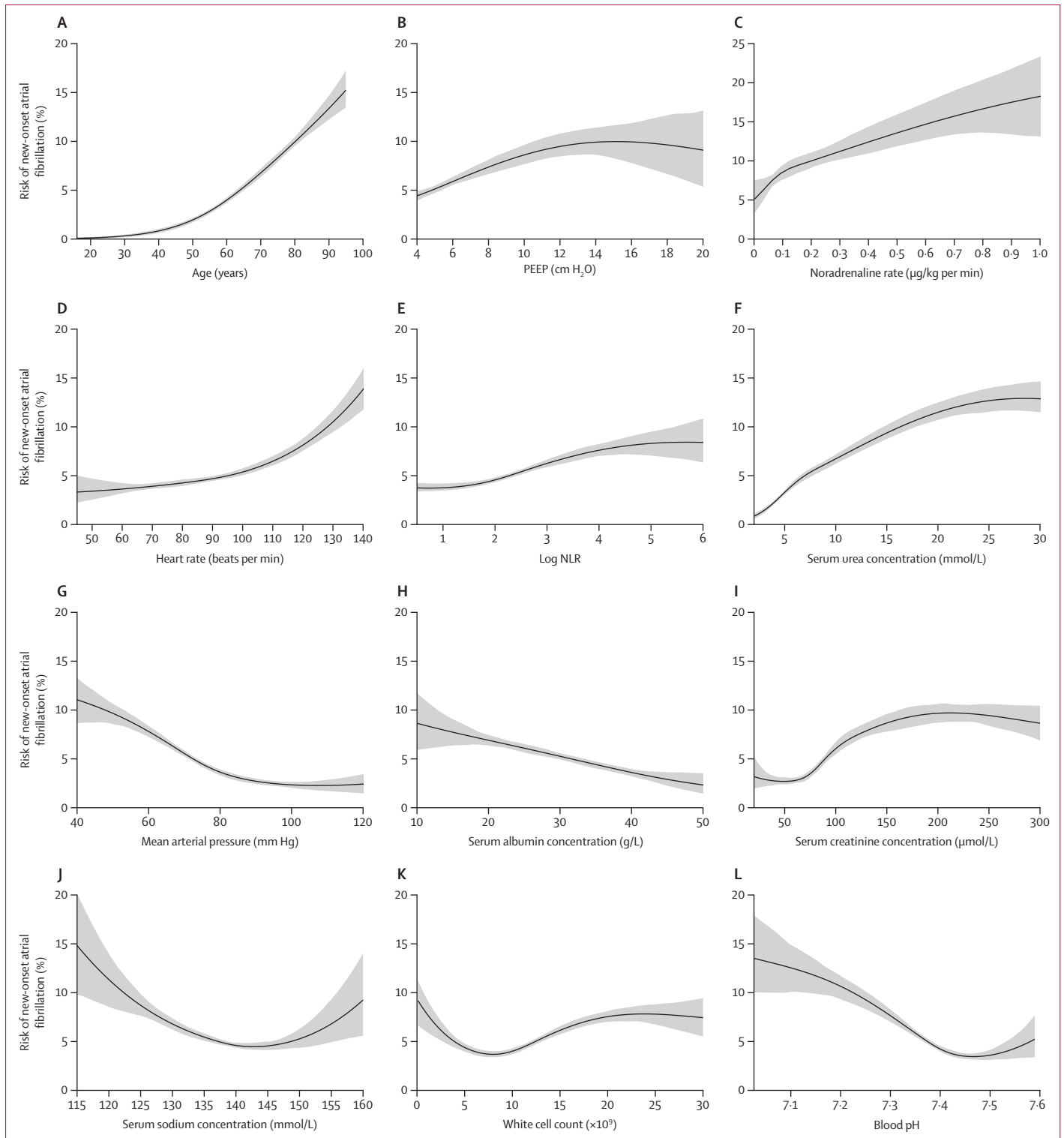


Figure 2: Univariable unadjusted associations between clinical variables and risk of new-onset atrial fibrillation occurring over the subsequent 24 h

Clinical variables assessed in the unadjusted analysis were age (A), PEEP (B), noradrenaline rate (C), heart rate (D), log NLR (E), serum urea concentration (F), mean arterial pressure (G), serum albumin concentration (H), serum creatinine concentration (I), serum sodium concentration (J), white cell count (K), and blood pH (L). Shaded areas represent 95% bootstrap CIs of predictions from models fitted to 1000 bootstrap samples of one value per patient, sampled throughout each patient's intensive care unit stay. Note that scales on y axes differ between plots. NLR=neutrophil-to-lymphocyte ratio. PEEP=positive end-expiratory pressure.

during ICU admission tended to be older, had longer ICU and hospital stays, and had higher ICU and hospital mortality compared with ICU patients who did not develop the condition (table; appendix p 23).

Data were incomplete for some variables. Missingness decreased during the initial hours of ICU admission as test results became available (appendix p 9). As a percentage of all 3-h time periods, the variables with the highest missingness after carrying forward were lymphocyte-to-neutrophil ratio (43·6%), because many white cell count results did not report a differential; serum lactate (24·1%); blood pH (22·3%); and fractional concentration of oxygen in inspired air (FiO₂; 11·6%). All other variables had a missingness of less than 5%. No outcomes were missing. Most laboratory tests were repeated daily, with serum magnesium newly measured on 95 766 (87·5%) patient days. Density plots comparing distributions of observed data with imputed data are shown in the appendix (p 10).

Of the candidate predictors, we found associations between both dichotomous predictors (appendix p 24) and continuous predictors (figure 2) and 24-h risk of new-onset atrial fibrillation. Serum magnesium concentration had a U-shaped relationship with risk in the unadjusted analysis. After accounting for confounding by renal impairment, the association between high magnesium and risk was no longer present; however, low magnesium (<0·70 mmol/L) remained associated with an increased risk (figure 3). Serum potassium concentration also had a U-shaped association with risk in the unadjusted analysis, which was less marked after adjustment; concentrations below 3·5 mmol/L were associated with increased risk of new-onset atrial fibrillation (figure 3).

The final model comprised 140 trees with a maximum depth of 4 and a learning rate of 0·07. Resampling hyperparameters were optimised at a column sampling proportion of 0·5 and a row sampling proportion of 0·7 for each tree. The METRIC-AF model achieved a mean internal-external cross-validation C statistic of 0·830 with ten predictors (95% CI 0·809–0·853). Performance improved only marginally with more predictors, to a maximum C statistic of 0·837 (95% CI 0·818–0·857) with all 47 candidate predictors (figure 4). Performance at each held-out ICU is shown in the appendix (p 14). Selected predictors comprised age, day of ICU stay, heart rate, FiO₂, maximum mean arterial pressure during stay, serum urea concentration, maximum serum urea concentration during ICU stay, presence of atrial ectopic beats, noradrenaline infusion rate, and presence of a central venous catheter. Discrimination performance was good (>0·80) when assessed in each of the seven held-out ICUs (C statistic 0·818–0·847; appendix p 24).

Partial dependence plots showed generally monotonically increasing average risk of new-onset atrial fibrillation associated with continuous variables in the METRIC-AF model, except for the negative associations observed with increasing day of ICU stay and mean arterial pressure. Overlaid individual conditional expectation plots showed

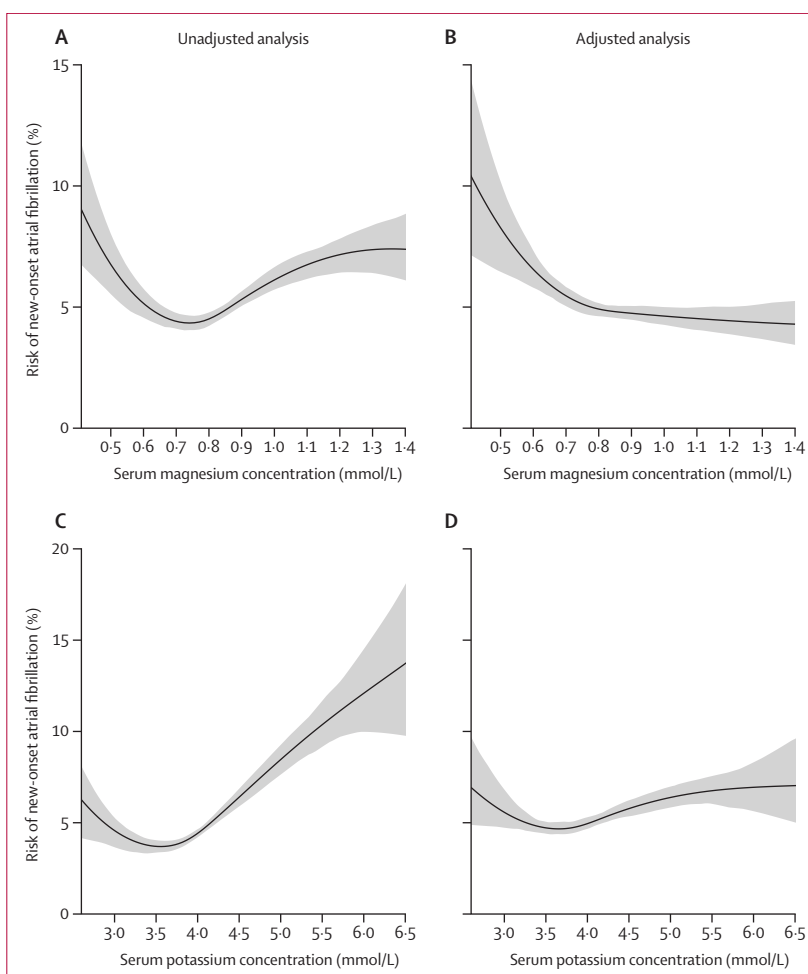


Figure 3: Associations between serum magnesium and potassium concentrations throughout each patient's ICU stay and risk of new-onset atrial fibrillation occurring over the subsequent 24 h

Associations between serum magnesium (A) and potassium (C) concentrations and risk of new-onset atrial fibrillation are provided for the unadjusted analysis. Associations between serum magnesium (B) and potassium (D) concentrations and risk are also provided for the analysis adjusted for renal function, assuming a serum urea concentration of 8·0 mmol/L and a serum creatinine concentration of 80 µmol/L. Shaded areas represent 95% bootstrap CIs of predictions from models fitted to 1000 bootstrap samples of one value per patient, sampled throughout each patient's ICU stay. ICU=intensive care unit.

variability between patients in both predicted baseline risk and changes in risk, with differences in predictor values (appendix p 15). In the Shapley analysis, age was the most important predictor of risk (appendix p 16).

After the comparator model was recalibrated, both models were well calibrated (appendix p 16). The METRIC-AF model had a significantly better discrimination than the comparator model (C statistic 0·812 [95% CI 0·805–0·822] vs 0·786 [0·778–0·801]; $p=0·0003$). Furthermore, the METRIC-AF model was found to have superior net benefit over various clinically important risk thresholds to the recalibrated comparator model (appendix p 17). At very low risk thresholds (<3%), the two models had similar net benefit. Both models were more useful than an intervene-on-all or intervene-on-none strategy. The model remained

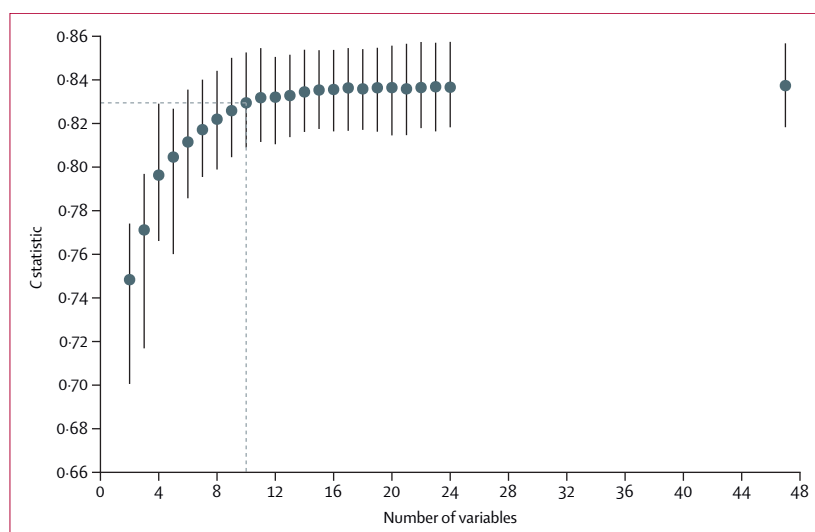


Figure 4: Internal-external cross-validation C statistic values by number of variables for the METRIC-AF model. Error bars represent 95% CIs. The dashed lines represent the model with the fewest predictors that achieved an area under the receiver operating characteristic within 1% of the model comprising all candidate predictors.

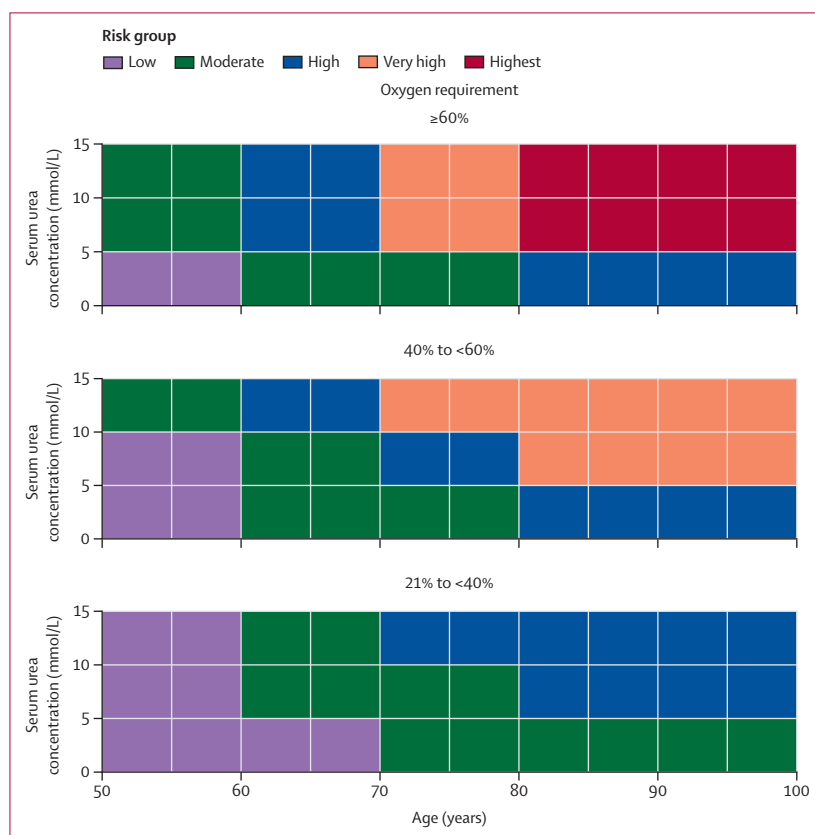


Figure 5: Simplified graphical tool stratifying intensive care unit patients by risk of new-onset atrial fibrillation. Stratification is based on three clinical predictors: age, serum urea concentration, and oxygen requirement. Patients younger than 50 years were classified as low risk.

well calibrated with good discrimination when assessed in subgroups stratified by sex, ethnicity, and age (appendix pp 18–19, 24).

External validation performance was assessed against multiple imputations of the missing variable of atrial ectopic beats. Model performance did not differ from the main findings or vary considerably across imputations (C statistic 0.813 [95% CI 0.807–0.818]). We performed a complete-case analysis of external validation, assessing model performance only among patients with no missing predictor values other than atrial ectopic beats. Of 29 833 eligible patients, 22 153 (74.3%) had complete data. 1856 (8.4%) patients had new-onset atrial fibrillation in this group. Model performance did not materially differ from the main findings (C statistic 0.804 [95% CI 0.785–0.808]).

A screenshot of the METRIC-AF app is shown in the appendix (p 20; table upload instructions not shown). The simplified graphical tool is shown in figure 5. Attributing risk of new-onset atrial fibrillation according to this tool in the external validation data achieved a C statistic of 0.727 (95% CI 0.716–0.739). The proportions of patients in different risk groups with new-onset atrial fibrillation over the subsequent 24 h and 7 days are shown in the appendix (pp 21, 25).

Discussion

Using data from seven ICUs across three centres in the UK and the USA, we developed a prediction model for new-onset atrial fibrillation in patients admitted to an ICU that incorporates objective and routinely collected clinical variables. The incidence of new-onset atrial fibrillation was consistent with review data.¹ The METRIC-AF model was well calibrated and showed good discrimination, with higher discriminative performance and superior net benefit over various clinically important risk thresholds than a published comparator model in the external validation dataset. We also developed a simple three-variable graphical tool using model-based predictions that stratifies patients by risk into five groups, including a large low-risk group unlikely to benefit from prophylaxis for new-onset atrial fibrillation.

We also explored individual predictor associations. After accounting for confounding, low serum magnesium concentration was associated with increased risk of new-onset atrial fibrillation. Although no causality can be attributed with this finding, this risk progressively increased with serum magnesium concentrations of less than 0.7 mmol/L. Future work could include causal inference studies to generate hypotheses to be examined in randomised controlled trials.³²

Our work has several strengths. Candidate predictors were defined with an international Delphi process, which reflects expert consensus and promotes transparency and relevance across diverse clinical settings. We developed and validated the METRIC-AF model using large, multicentre databases and robust methods to mitigate overfitting. However, this work should be interpreted in the context of its limitations. The development dataset comprised ICUs from the UK and the USA only, including both district general and tertiary centres. Although this diversity

enhances the generalisability of our model, it might not fully capture site-specific or country-specific variations in ICU populations, particularly in regions not included in the study. Furthermore, given the timeframe of the data, changes in demographics and clinical practices could influence the generalisability of our findings. The outcome (new-onset atrial fibrillation) was not independently verified; however, documentation of atrial fibrillation in an earlier MIMIC database was shown to be accurate for identifying the onset of atrial fibrillation to within 1 h of true onset in an independent review of electrocardiogram waveforms.³³ Although the gain statistics used during predictor selection implicitly captured some information on interactions through sequential splits in trees, they did not comprehensively account for all interactions between features. External validation performance was limited by the absence of data for one predictor (atrial ectopy). However, the sensitivity analysis showed consistent performance across multiple imputations of this missing predictor. The performance of the METRIC-AF model might exceed that reported here if tested in data that include ectopy. The METRIC-AF model slightly underestimated the overall risk of new-onset atrial fibrillation, possibly accounted for by the higher proportion of patients with the outcome in the external validation data than in the development data. External validation was performed with data from multiple UK centres, strengthening its generalisability within the UK context. However, external validation in only UK settings could limit the interpretation of our validation results for international populations. Future studies should seek to validate the model with international data. Although the recalibrated European comparator model was quite well calibrated to validation data from the UK, it was previously shown to be very poorly calibrated to validation data from the USA.³⁴

Existing published models have limitations that prevent their widespread adoption. One model developed with data from two ICUs in the Netherlands¹⁰ was subjected to an external validation study with data from ICUs in the USA, reporting a C statistic of 0.598 with poor calibration.³⁴ A 2022 model for predicting new-onset atrial fibrillation leveraged a large cohort (16 144 patients) from one centre in Amsterdam comprising a mixed ICU and a high dependency unit.¹¹ However, the model included predictors that are not routinely measured, such as height of the ST segment, and did not include an analysis of calibration or external validation. A subsequent robust study that used the same data showed promising performance, but required mean values over a patient's stay in the ICU, limiting clinical utility.³⁵ A machine learning model developed with MIMIC-IV data and validated on MIMIC-III data performed less well than the METRIC-AF model (C statistic 0.769 [95% CI 0.756–0.782]), despite the same source of development and validation data.³⁶

We found that, below a concentration of 0.7 mmol/L, further decreases in serum magnesium concentration are associated with an increasing risk of new-onset atrial

fibrillation. This observation is supported by the findings of a randomised study showing that prophylactic supplementation with magnesium lowered risk of postoperative atrial fibrillation following cardiac surgery.³⁷ We found no evidence to support the common practice of targeting high to normal serum concentrations of magnesium in patients at risk of atrial arrhythmias in general ICUs.³⁸

The ten selected model predictors are highly plausible and include key indicators of organ failure. Uraemic toxins might also promote atrial fibrillation by increasing oxidative stress.³⁹ The presence of atrial ectopic beats is not found in other ICU scoring systems and might represent a specific risk of atrial fibrillation, rather than a surrogate of illness severity, with atrial ectopy initiating episodes of atrial fibrillation.⁴⁰ Including the day of admission as a predictor in the METRIC-AF model is consistent with the published comparator model, with both models finding an increased risk of new-onset atrial fibrillation earlier in the ICU stay.¹⁰

Although the METRIC-AF model has undergone external validation, its predictive performance needs further evaluation. A function in the predictive tool allows the appending of predicted risks to any uploaded table, encouraging external validation. The METRIC-AF model or its companion graphical tool could identify patients at sufficient risk of new-onset atrial fibrillation for recruitment in prevention studies, allowing the progression of evidence in this important area. Taken with previous work, our finding that hypomagnesaemia was associated with new-onset atrial fibrillation suggests that serum magnesium targets should be investigated in randomised trials of patients at high new-onset atrial fibrillation risk in general ICUs.

Overall, the METRIC-AF model outperformed an existing model for predicting new-onset atrial fibrillation in a large external dataset of patients in ICU. Serum magnesium concentrations of less than 0.7 mmol/L were associated with an increased risk of the condition. Future work should aim to further validate the METRIC-AF model in different patient populations. Identifying patients at high risk could enable targeted trials of preventive interventions to improve clinical outcomes.

Contributors

JPB led the conceptualisation of the study, secured funding, curated and verified the raw data, and created the figures. JPB also wrote the original draft of the manuscript and serves as the guarantor of the paper. RH and OR provided resources, and curated and verified the raw data. SG assisted with data curation and the formal analysis. LK contributed to data curation and to the review and editing of the manuscript. DC and GSC contributed to the methodology and to the review and editing of the manuscript. PJW was responsible for securing funding, providing resources, and supervising the project. JPB directly accessed and verified the underlying data. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

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Data sharing

The MIMIC-IV dataset is openly available and can be accessed at <https://mimic.physionet.org/>. The PICRAM dataset is not publicly available and cannot be shared. A link to an online repository, including sample code for the model development used in the study, is provided on GitHub.

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For the online repository see
https://github.com/JonNDCN/metric_af

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